

ANSWER KEY — Integration Lesson 14 Test: The Average Value of a Function

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Find the average value of the constant function $f(x) = 9$ on the interval $[0, 5]$.

Work: prediction: a function that is always 9 must average 9. Formula: $\int_0^5 9 \, dx = 9x$ from 0 to 5 = 45; average = $45 \div (5 - 0) = 9$. **Answer: 9.**

2. Find the average value of $f(x) = 3x$ on $[0, 4]$. Check with the endpoint average.

Work: $\int_0^4 3x \, dx = (3/2)x^2$ from 0 to 4 = 24. Average = $24 \div 4 = 6$. Endpoint check: $f(0) = 0$, $f(4) = 12$, $(0 + 12)/2 = 6$ ✓ — allowed because $3x$ is a straight line (only lines climb at a steady pace, so their two endpoints tell the whole story). **Answer: 6.**

3. Find the average value of $f(x) = x^2$ on $[0, 12]$.

Work: $\int_0^{12} x^2 \, dx = x^3/3$ from 0 to 12 = $1728/3 = 576$. Average = $576 \div 12 = 48$. (Plausibility: between $f(0) = 0$ and $f(12) = 144$, and below the midpoint 72 since x^2 lingers low.) **Answer: 48.**

4. The average value of f on $[0, 6]$ is 8. What is $\int_0^6 f(x) \, dx$?

Work: run the formula backwards: total = average $\times$ length = $8 \times 6 = 48$. (Picture: settled water at height 8 across width 6 $\rightarrow$ area 48.) **Answer: 48.**

5. Find the average value of $f(x) = x + 5$ on $[0, 8]$.

Work: $\int_0^8 (x + 5) \, dx = x^2/2 + 5x$ from 0 to 8 = $32 + 40 = 72$. Average = $72 \div 8 = 9$. Endpoint check: $(f(0) + f(8))/2 = (5 + 13)/2 = 9$ ✓ (straight line). **Answer: 9.**

6. $T(t) = 60 + 2t$ °F. Average temperature from $t = 0$ to $t = 10$?

Work: $\int_0^{10} (60 + 2t) \, dt = 60t + t^2$ from 0 to 10 = $600 + 100 = 700$. Average = $700 \div 10 = 70$. Check: line from 60 to 80, midpoint $(60 + 80)/2 = 70$ ✓. **Answer: 70°F.**

7. $v(t) = 6t^2$ m/s from $t = 0$ to $t = 2$. (a) Total distance? (b) Average velocity?

Work: (a) $\int_0^2 6t^2 \, dt = 2t^3$ from 0 to 2 = 16 meters. (b) Average velocity = $16 \div 2 = 8$ m/s. (The endpoint trick would give $(0 + 24)/2 = 12$ — wrong, because $6t^2$ is a curve that spends most of the time slow.) **Answer: (a) 16 meters; (b) 8 m/s.**

8. The average value of $f(x) = x^2$ on $[0, c]$ is 3. Find c .

Work: $\int_0^c x^2 dx = c^3/3$, so average $= (c^3/3) \div c = c^2/3$. Set $c^2/3 = 3 \rightarrow c^2 = 9 \rightarrow c = 3$ (positive root). Check: on $[0, 3]$ total $= 9$, average $= 9/3 = 3$ ✓. **Answer: $c = 3$.**

9. Why does the endpoint method overshoot for x^2 on $[0, 6]$ (18 vs. the true 12), and what kind of function does it work for?

Work: the endpoint method assumes the function climbs at a steady pace between its two ends. But x^2 hugs low values for most of the interval (at $x = 3$, halfway across, it has only reached 9 out of 36) and sprints up at the end. An average weights values by how long the function spends near them, so the true average leans low: $12 < 18$. The endpoint method is guaranteed only for **straight lines**. **Answer: x^2 spends most of its time at small values; endpoint-averaging works only for straight lines.**

10. $v(t) = t^2 + 2t$ m/s on $[0, 3]$. (a) Net change in position? (b) Average velocity? (c) Consistency check.

Work: (a) $\int_0^3 (t^2 + 2t) dt = t^3/3 + t^2$ from 0 to 3 $= 9 + 9 = 18$ meters. (b) Average velocity $= 18 \div 3 = 6$ m/s. (c) Constant 6 m/s for 3 s $\rightarrow 6 \times 3 = 18$ meters — same net change ✓. **Answer: (a) 18 meters; (b) 6 m/s; (c) $6 \times 3 = 18$ ✓.**